

ORF309/MAT380

**Barnoulli Processes II: Arrival
Times & Geometric Distribution**

(pp.42-47)

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Agenda for today

- *All about “arrivals of successes”, today.....*

Recap from previous lecture

- The DRV X_i that takes values in $\{0,1\}$, with $P\{X_i = 1\} = p$ and $P\{X_i = 0\} = q$ and $p + q = 1$ is called **Bernoulli RV**
- A sequence of n Bernoulli RVs added together is call **Bernoulli Process** and **counts the number of successes in n trials**
- $S_n = X_1 + X_2 + \dots + X_n$ (S_n is also a RV)

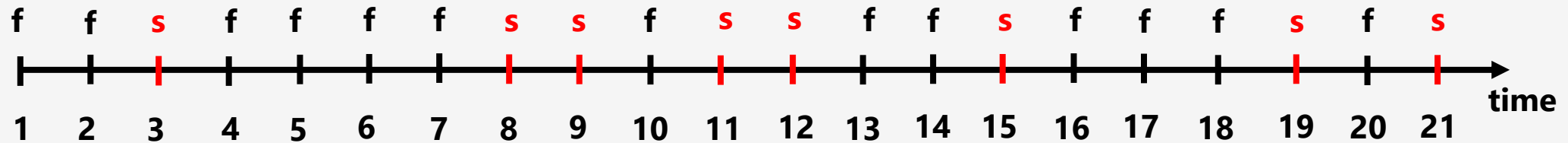
$$P\{S_n = k\} = \binom{n}{k} p^k q^{n-k} \quad \text{and} \quad S_n \sim \text{Binom}(n, p)$$

Time (or Trial) of the k -th Success – Negative Binomial Distribution

Time of the k -th Success

- Consider the DRV: T_k = "the time (or trial) of the k -th success"
 - We will compute its probability distribution: $P\{T_k = n\}$ for $n \geq 1$

- Consider the following **example**:



- $T_1 = 3, T_2 = 8, T_3 = 9, T_4 = 11, \dots$

Note: Here we consider the time as discrete. So, the time increases when we perform a new trial

Time of the k -th Success

- Let's look closer at the event $A = \{T_k = n\}$
- **Can I write event A differently?** (**Hint**: think of X_n and S_{n-1})

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Time of the k -th Success

- Let's look closer at the event $A = \{T_k = n\}$
- **Can I write event A differently?** (**Hint**: think of X_n and S_{n-1})
 - $A = \{X_n = 1, S_{n-1} = k - 1\}$

- Then we can easily compute the probability $P\{T_k = n\}$:

$$P\{T_k = n\} = P\{X_n = 1, S_{n-1} = k\} \stackrel{\perp\!\!\!\perp}{=} P\{X_n = 1\}P\{S_{n-1} = k - 1\} = p \binom{n-1}{k-1} p^{k-1} q^{n-k}$$

No further simplification is possible!

Negative Binomial Distribution

- **Definition:** A DRV Z with values $k \leq n < \infty$ such that

$$P\{Z = n\} = \binom{n-1}{k-1} p^k q^{n-k}$$

is said to have **Negative Binomial Distribution** with parameters k and p

- **Notation:** $Z \sim \text{NegBinom}(k, p)$

Time (or Trial) until the 1st Success –
Geometric Distribution

Time until the 1st success – Geometric Distribution

- **Special case of Negative Binomial (for $k = 1$):**
- Let's consider: $T_1 = \text{"the time until we get the 1st success"}$

$$P\{T_1 = n\} = \binom{n-1}{1-1} p^1 q^{n-1} = pq^{n-1}$$

- **It is very intuitive:** to get the 1st success in n trials we must get $n - 1$ failures
- **Definition:** A DRV that takes values in $\{1, 2, 3, \dots\}$ such that $P\{Z = n\} = pq^{n-1}$ with $p \in [0, 1]$, and $q = 1 - p$ is said have the **Geometric distribution** with parameter p
 - **Notation:** $Z \sim \text{Geom}(p)$
 - **Remark:** this is the distribution of the 1st success

Can we fail for ever ?

When can we fail for ever?

- **Question:** Can we fail for ever?? → **How can I write this in terms of T_1 ??**

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- Case 1: if $p = 0$ → probability of success is 0 (Tough luck! Don't fight it!)
 - $P\{T_1 = \infty\} = 1$
- Case 2: if $0 < p \leq 1$ → there is some hope...
 - $P\{T_1 = \infty\} =$ But how can I write the event $\{T_1 = \infty\}$ in terms of the $X_1, X_2, X_3, \dots$

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 - $P\{T_1 = \infty\} = P\{X_1 = 0, X_2 = 0, X_3 = 0, \dots\} \stackrel{II}{=} qqq \dots = q^\infty = 0 \Rightarrow P\{T_1 < \infty\} = ??$

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 - Hence if there is even a very small probability to succeed, we will succeed. **So, keep trying!** (and be always optimistic!)

Murphy's law

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- **Case 2 can be thought of as **Murphy's law****, if we consider p as the probability of **failure**. **The message is:** if failure can happen, then it will happen!
- **Another way of looking at it:** Since $P\{T_1 = \infty\} = 0$ when $p > 0$, then we should have $P\{T_1 < \infty\} = 1$ (because $\{T_1 < \infty\} = \{T_1 = \infty\}^c$).
 - Then we have: $1 = P\{T_1 < \infty\} = P\{T_1 = 1 \text{ OR } T_1 = 2 \text{ OR } T_1 = 3 \text{ OR } \dots\} =$

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 $= P\{T_1 = 1\} + P\{T_1 = 2\} + P\{T_1 = 3\} \dots = \sum_{k=1}^{\infty} pq^{k-1} \Rightarrow p \sum_{k=1}^{\infty} q^{k-1} = 1 \Rightarrow \sum_{k=1}^{\infty} q^{k-1} = \frac{1}{1-q}$ Geometric series

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Geometric series

- **Remark:** We just proved that the **sum of Geometric series** is equal to 1
(without using any Calculus!)

Expectation of T_1

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- Recall that $T_1 \sim \text{Geom}(p)$, with $p > 0$.

$$E[T_1] = \sum_{k=1}^{\infty} k P\{T_1 = k\} = \sum_{k=1}^{\infty} k p q^{k-1} = p \sum_{k=1}^{\infty} k q^{k-1} = ??$$

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Expectation of T_1

- **Let's interpret the result:**
 - If you know the probability of success (p), then **you will get your first success, on average, after $1/p$ trials**
 - The **smaller** the probability of success the **larger** the number of trials you have to perform
 - If $p = 0.001$ then $E[T_1] = 1000$
 - Similar interpretations hold for the appearance of the **first failure**

Joint Distribution between Successes

Joint probability distribution between successes

- **QUESTION:** *How long we need to wait between successes?*
- Let's compute the **probability** that the 1st success comes after 3 trials, the next success comes after 5 trials, the next success comes after 1 trial, the next success comes after 6 trials, the next success comes after 3 trials, the next success comes after 3 trials.
- How can I express this probability in terms of $T_1, T_2, T_3, \dots$??

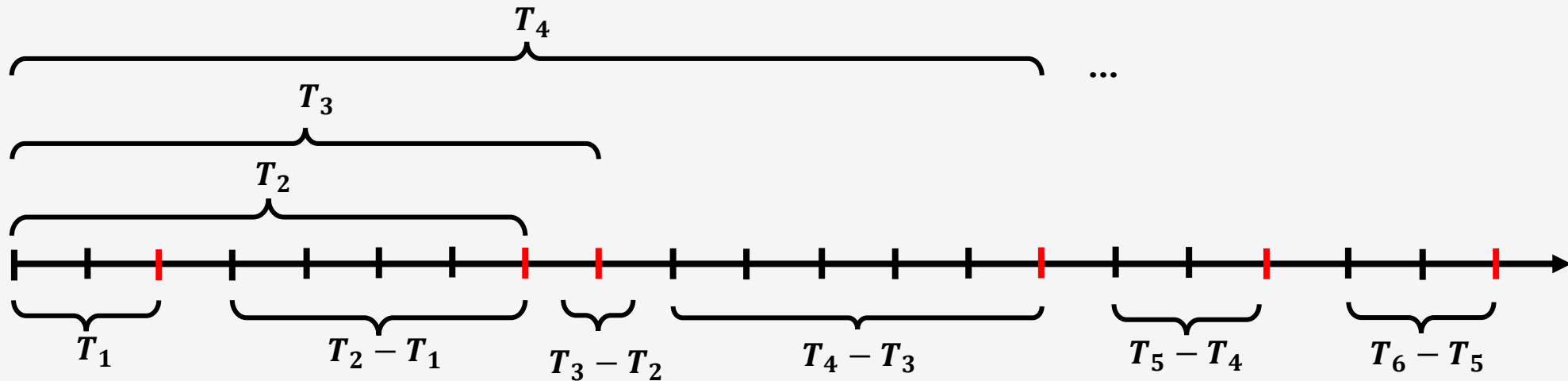
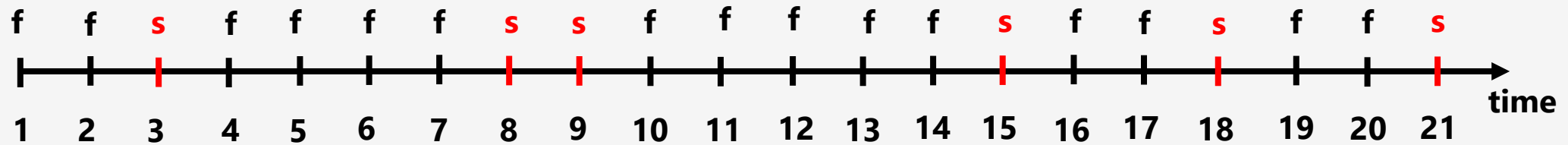
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- How can I express this probability in terms of $T_1, T_2, T_3, \dots$??

$$P\{T_1 = 3, T_2 - T_1 = 5, T_3 - T_2 = 1, T_4 - T_3 = 6, T_5 - T_4 = 3, T_6 - T_5 = 3\} = ??$$

Joint probability distribution between successes

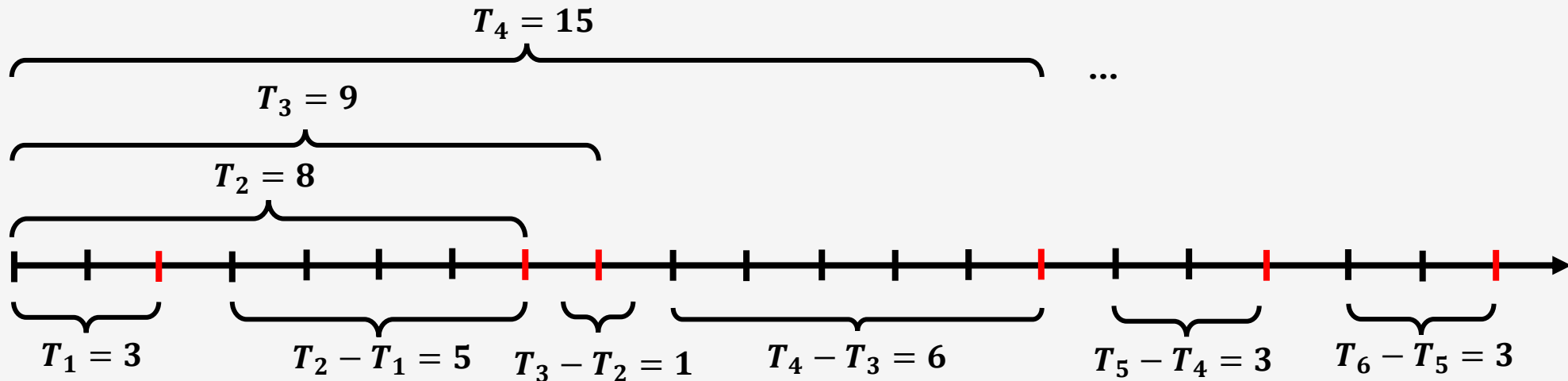
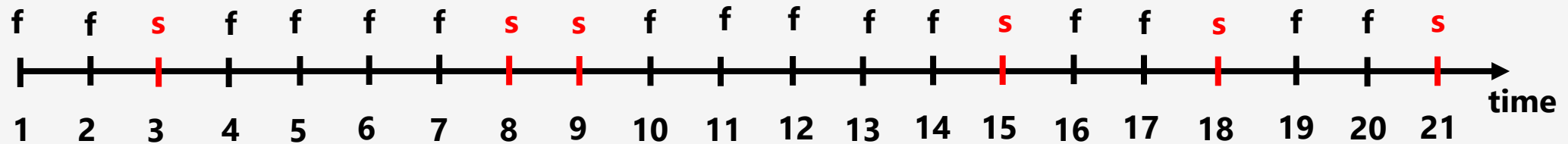
- Illustrations:



Note: each random variable $T_{i+1} - T_i$ includes the **success** at its end

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$$\begin{aligned} P\{T_1 = 3, T_2 - T_1 = 5, T_3 - T_2 = 1, T_4 - T_3 = 6, T_5 - T_4 = 3, T_6 - T_5 = 3\} = \\ = P\{ff\textcolor{red}{s}, fffffs, \textcolor{red}{s}, ffffffs, ffs, ffs\} = q^2p \ q^4p \ p \ q^5p \ q^2p \ q^2p = \end{aligned}$$

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- **Remark:** The events $\{T_{i+1} - T_i = k\}$ and $\{T_1 = k\}$ have the **same distribution** which is a **Geometric distribution** with parameter p
 - Think of resetting your clock every time a success happens.

Telescopic sum & $E[T_k]$

Telescopic sum and $E[T_k]$

- Recall that the $T_k \sim \text{NegBinom}(k, p)$
- We use the telescopic sum to write T_k as a sum of Geometric RVs

$$T_k = T_k - T_{k-1} + T_{k-1} - T_{k-2} + T_{k-2} - \cdots - T_2 + T_2 - T_1 + T_1$$

- We have: $T_i - T_{i-1} \sim \text{Geom}(p)$ and $T_i - T_{i-1} \perp\!\!\!\perp T_j - T_{j-1}, \forall i, j$
- Remark:** We showed a similar result for the Binomial distribution
 - The Binomial distribution is the sum of Bernoulli distributions

Let's now compute $E[T_k]$

- **Question:** What is the expected time at which the k -th success will happen??
- You can compute $E[T_k]$ using the definition and the distribution of $T_k \sim \text{NegBin}(k, p)$ **BUT it will be NOT very easy...**

Let's now compute $E[T_k]$

- **Question:** What is the expected time at which the k -th success will happen??
- You can compute $E[T_k]$ using the definition and the distribution of $T_k \sim \text{NegBin}(k, p)$ **BUT it will be NOT very easy...**
- You can use a **shortcut** $\rightarrow$ use **Telescopic sum** & $T_i - T_{i-1} \sim \text{Geom}(p)$ & $\perp\!\!\!\perp$

$$E[T_k] = E[T_k - T_{k-1}] + E[T_{k-1} - T_{k-2}] + \cdots + E[T_2 - T_1] + E[T_1] = \frac{k}{p}$$

Can we succeed infinitely often ?

Can you succeed infinitely often?

- **Question:** if the probability of success $p > 0$, will we succeed infinitely often??

$$P\{\text{success occurs infinitely often}\} = P\{T_1 < \infty, T_2 - T_1 < \infty, T_3 - T_2 < \infty, \dots\}$$

$\prod$

$$= P\{T_1 < \infty\}P\{T_1 < \infty\}P\{T_1 < \infty\} \dots = 1 \times 1 \times 1 \times \dots = 1$$

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$$= P\{T_1 < \infty\}P\{T_1 < \infty\}P\{T_1 < \infty\} \dots = 1 \times 1 \times 1 \times \dots = 1$$

- **Strong Murphy's law:** if something can happen, it will happen infinitely many times.

Practice exercises/examples

(may not be covered during lectures but they are offered for your practice)

Exercise 1:

- In the **Venture Capital (VC) world** investors put money into early-stage startups. Historical evidence suggests that most of these startups will fail only very few will become Unicorns (those that will reach \$1B valuation).
- An investment is considered successful for the VC firm only if the startup becomes **unicorn**. Unicorns can return 60x the initial investment. A start up has 1% probability of becoming a unicorn.
- The next best successful investment is when the startup performs a "**successful exit**". In this case you get 10x your initial investment. A start up has 10% probability for a "successful exit".
- **Modeling assumption:** All startups are independent of each others.
- **The goal:** The VC firm has invested in $n = 50$ startups. To ensure its profitability the VC firm to meet at least one of the two goals:
 - **Goal 1:** Have at least one unicorn; **Goal 2:** Have at least six "successful exits"
 - Find the **probability** of Goal 1 and the probability of Goal 2.
 - If you were the owner of the VC firm, which goal you would rather follow? (your personal choice!)

Solution: (try solving it without reading this first)

Solution

- Define: X_i = "the i -th startup becomes unicorn" $\in \{0,1\}$, Y_i = "the i -th startup becomes successful exit" $\in \{0,1\}$.
- Define the probabilities: $P(X_i) = 0.01 = p_U$ and $P\{Y_i\} = 0.1 = p_S$ for $i = 1, 2, \dots, 50$
- Modeling assumption: Startup outcomes are independent, that is, (X_i, Y_i) are i.i.d. for all $i = 1, \dots, 50$
- Let's define: U_n = "# of unicorns" and S_n = "# of successful exits in n startups" with $n = 50$.
- $U_{50} = X_1 + \dots + X_{50}$ and $S_{50} = Y_1 + \dots + Y_{50}$
- Then $U_{50} \sim \text{Binom}(50, 0.01)$ and $S_{50} \sim \text{Binom}(50, 0.1)$
- $P\{U_{50} \geq 1\} = 1 - P\{U_{50} = 0\} = 1 - \binom{50}{0} 0.01^0 0.99^{50} = 1 - 0.99^{50} = 0.395$ (use of complement event!)
- $P\{S_{50} \geq 6\} = 1 - P\{S_{50} \leq 5\} = 1 - \sum_{k=0}^5 \binom{50}{k} 0.1^k 0.9^{50-k} = 0.3837$ (use of complement event!)
- The probabilities are very close to each other, so the decision is a personal one.

Exercise 2

- A VC firm invest sequentially in a number startups, one per quarter. Each startup independently results in
 - a successful exit with probability 10%
 - A failure with probability 90%
- The VC firm decides to stop investing as soon as it achieves its 5th successful exit startup (because at that point the VC is considered viable).
- Your tasks:
 - a) What is the probability that the fund becomes viable on or before the 40th investment trial?
 - b) What is the probability that the 40th investment trial is the one that delivers the 5th successful exit?

Solution: (try solving it without reading this first)

Exercise 2

- Let X_i = "the i -th investment will become a successful exit", and $P\{X_i\} = 0.1$, for all $i = 1, 2, \dots$
- All X_i are independent
- The stopping criterion for the VC firm: Stops investing when it has $k = 5$ successful exits

a) We are looking for $P\{T_5 \leq 40\}$

Define the DRV: X = "there at least 5 successful exits among the first 40 quarters (investments)"

Then $X \sim \text{Binom}(40, 0.1)$ and $P\{T_5 \leq 40\} = P\{X \geq 5\} = 1 - P\{X \leq 4\} = 1 - \sum_{i=0}^4 \binom{40}{i} 0.1^i 0.9^{40-i} = \dots$

b) Define the DRV: T_k = "the # of quarters (i.e., # of investments) until the k -th successful exit"

- T_k must follow Negative Binomial distribution, $T_k \sim \text{NegBinom}(k, p)$, where $k = 5, p = 0.1$
- Hence, $P\{T_5 = 40\} = \binom{39}{4} p^5 (1-p)^{35} = \binom{39}{4} 0.1^5 0.9^{35} = \dots$